

Compensation of the Transmission Errors in Electrically Preloaded Rack-and-Pinion Drives

Lukas Steinle, Valentin Leipe, Armin Lechler, and Alexander Verl
Institute for Control Engineering of Machine Tools and Manufacturing Units
University of Stuttgart
Stuttgart, Germany
lukas.steinle@isw.uni-stuttgart.de

Abstract—Large machine tools with long feed axes commonly utilize electrically preloaded rack-and-pinion drives, where two redundant drives on one axis are preloaded against each other by a dedicated control loop. Manufacturing- and assembly-related tolerances and elastic deformations lead to transmission errors in the drivetrains, that limit the achievable accuracy. Due to the limited bandwidth of the position control, these cannot be fully suppressed despite the use of linear measuring systems. In this paper, the superposition of the individual errors of the two drivetrains of electrically preloaded systems and the transfer function to the resulting path errors are theoretically derived in a first step. In addition, the negative effect of transmission errors on the preload control is investigated. To address the issue, an error compensation for the position and preload control loops is implemented. The basis for this are models of the state-dependent transmission errors utilizing a two-stage machine learning approach. The achievable reduction of path errors is validated in experimental investigations on a test bench and averages 57.8 % for the measurements conducted.

Index Terms—Machine tools, Servosystems, Position control, Transmission errors, Error compensation

I. INTRODUCTION

In machine tools, the installed feed drives are decisive for the achievable dynamics as well as the positioning and path accuracy [1]. For machines with long travel distances and high feed and acceleration forces, rack-and-pinion drives (RPDs) are the preferred choice [2]. Their main advantage is the potential to realize arbitrarily long travel distances by lining up stationary rack elements without compromising the dynamics or rigidity of the drive [3]. However, the limited accuracy represents a major drawback of RPDs. This is due to tolerance-induced transmission errors (TEs) in the drivetrains, which lead to deviations in the linear position of the feed axis [4], [5]. To minimize the negative effects of backlash and thus increase accuracy, in industrial applications two redundant drives are installed on one axis, with the pinions statically preloaded against each other by a preload control [6]. This electrical preloading offers the advantage of a controlled and adjustable preload torque, which makes it the preferred choice for large machine tools [2].

However, the TEs still have a significant influence on the path accuracy. The relationship between the rotation of

the motor and the translation of the feed axis, which is generally assumed to be ideal, is subject to variations in reality. Manufacturing and assembly-related kinematic errors of the rack, pinion and gearbox as well as the nonlinear stiffness of the gearing cause deviations from the ideal transmission ratio. A comprehensive analysis of these TEs was conducted on single drive RPDs [4]. The study revealed direction-dependent characteristics and significant load-dependent alterations of the observed TEs. To negate these deviations, linear measuring systems are installed in applications with high demands to feed back the linear position to the drive control. However, due to the limited bandwidth of the position control, periodic components in particular cannot be completely suppressed [7], [8]. Load- and velocity-dependent residual errors remain, which impair the machining quality and limit the applicability of machines with RPDs.

In previous work, the implications of TEs for the path accuracy of single drive RPDs were analyzed in detail and the theoretical background was derived and validated [4], [7]. A machine learning approach for modeling the individual and state-dependent transmission errors of one drive was developed and an error compensation was integrated in the drive control. The achievable increase in path accuracy was experimentally evaluated and quantified in [7] and [9]. However, this previous work has focused explicitly on single drive RPDs. For electrically preloaded systems with two redundant drives, the path errors result from a combination of the individual errors of the two drivetrains. The error composition as well as the theoretical background have not yet been investigated. In addition to this, an interference of the TEs with the preload control has been described in the literature but has not yet been analyzed [10]. Accordingly, the error compensation approach developed for single drives cannot directly be transferred to the more complex control structure for electrically preloaded systems.

Given the widespread use of electrically preloaded RPDs, this publication aims to close this gap. Therefore, the fundamentals of the control of electrically preloaded systems are first explained in Sec. II, before the characteristics of the TEs of RPDs are outlined in Sec. III. Subsequently, the implications of the TEs on the path accuracy and preload control are investigated in Sec. IV. Building on the knowledge gained, Sec. V develops a TE compensation for the position and

This work was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation), Project No. 521502188. The authors gratefully acknowledge the support by DFG.

preload control of electrically preloaded RPDs based on the method presented in [7]. Finally, an experimental validation combined with a quantification of the achieved improvement in path accuracy is carried out on the test system in Sect. VI.

All measurement data in this publication, including variables that are not displayed and supplementary data, can be accessed in [11].

II. CONTROL OF ELECTRICALLY PRELOADED RACK-AND-PINION DRIVES

For preloaded RPDs, the axis motion is provided by two pinions that are preloaded against each other by a differential torque, that prevents both drives from undergoing backlash at the same time [12]. This ensures that force can always be exerted on the machine table and the system therefore remains controllable, thus significantly improving accuracy. The preload torque is specified as a proportion of the nominal torque of the drive motors τ_n and typically ranges between 10 % and 30 % [13]. Higher preloading increases the stiffness of the system, but also increases energy consumption [6], [14]. In the state of the art, the preload torque is mostly constant during operation, however, there are approaches for dynamic adaptation based on the operating state, in order to improve performance and energy efficiency [12], [15], [16].

The industrial standard for the position control of feed drives is a cascade control, consisting of three nested control loops for the drive torque, velocity and position [1]. For electrically preloaded RPDs, this structure is extended by a torque balancing control [12], [13]. The respective control structure is shown in Fig. 1, whereas the torque control loops are not depicted for simplification. The mechanical subsystem can be represented as a three-mass oscillator consisting of the inertia of the two drives and the table [14]. The target torques $\tau_{d,1/2}$ for the two drives are determined by two dedicated velocity controllers based on the current angular velocity of the motors $\dot{\theta}_{M,1/2}$. They are implemented as proportional-integral (PI) controllers, with the gain K_p and the integrator time constant T_n . The preload torque $\tau_{pl,d}$ is applied by a dedicated control loop based on the difference between the

TABLE I: Key system parameters.

Parameter	Symbol	Value	Unit
Gearing module	m	4	mm
Pinion pitch diameter	d_p	84.882	mm
Number of pinion teeth	n_t	20	-
Gearbox ratio	i_{PG}	16	-
Vel. control gain	K_p	6.2	Nm s/rad
Vel. control integrator time constant	T_n	3.05	ms
Nominal motor torque	τ_n	10	Nm
Preload torque	$\tau_{pl,d}$	1.25	Nm
Preload control gain	K_{pl}	0.3	rad/Nm s
Pos. control gain	K_v	33	1/s

target torques of the drives. The output is an additive angular velocity offset $\dot{\theta}_p$ between the two drives. This controller is typically implemented as a P-controller with the gain K_{pl} . The higher-level position controller refers to the preloaded drive as a whole, with the position error e_x representing the difference between the target position x_d and the table position x_T . It is implemented as a proportional (P) controller with the gain K_v . In combination with the commonly applied velocity feedforward, the target velocity v_d is obtained. As the position control refers to the linear position, whereas the velocity controllers refer to the angular velocity of the motors, the angular target velocity $\dot{\theta}_d$ is calculated using the nominal transmission ratio i_n of the drivetrains.

III. TRANSMISSION ERRORS OF RACK-AND-PINION DRIVES

The basic characteristics of the TEs of RPDs were studied in detail by the authors in the context of single drive systems [4], [7]. The key findings are outlined in this section through measurements on an electrically preloaded system, which is first introduced.

A. Experimental setup

All measurements in this publication are conducted on the test bench shown in Fig. 2. The setup is a single axis with electrically preloaded RPDs, which uses industry standard components and was described in detail in a previous publication [16]. It features two identical RPDs with helical gearing that are driven by servomotors via two-stage planetary gearboxes. The key system parameters are listed in Table I. The rotary encoders in both motors as well as a linear measuring system integrated into the guideways are utilized for position measurement. A redundant linear direct drive (LDD) is mounted between the guideways. It is operated in force control and acts as a load machine to allow the application of disturbance forces to the drive system. The parameterization of the drive control, which is also summarized in Table I, follows established approaches. The velocity controllers were both tuned identically in accordance with the standard procedure for Siemens drive controllers [17]. The parameterization of the torque balancing control follows the procedure described in [10] with a preload torque of 12.5 % (1.25 Nm). The position control was subsequently tuned to be free of overshoot (aperiodic loop) [13].

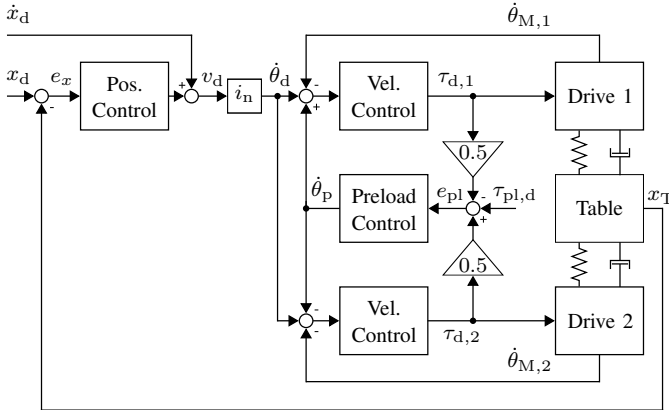


Fig. 1: Drive control with torque balancing control for electrically preloaded RPDs, based on [12], [13].

B. Identification of the transmission errors

The transmission error

$$TE = x_T - x_M = x_T - \frac{\theta_M}{i_n} = x_T - \frac{\theta_M d_P}{2i_{PG}} \quad (1)$$

of a RPD denotes the difference between the table position and the linear motor position x_M , which is derived from the angular motor position through the nominal transmission ratio of the drive. The TEs are position- and load-dependent [4]. For electrically preloaded systems, an identification of the errors can be performed by varying the preload torque. Therefore, the axis is traversed repeatedly at a low velocity of 10 mm/s to exclude dynamic effects, while the preload torque is varied between the individual runs. The individual cycles are then assigned to the drive whose direction of transmitted force corresponds to the direction of the motion. Fig. 3 shows the obtained transmission error data for drive 2 of the test bench. The error curves in the lower half of the diagram represent a force transmission in the positive direction, while those in the upper half correspond to the negative direction. The highlighted area between the no-load error curves for both directions reflects the position-dependent backlash of the drivetrain.

The error curves consist of three main components [4]. Firstly, there is a drift of the table position, that is caused by pitch deviations in the rack elements and assembly tolerances, which lead to variations of the center distance and height alignment between the pinion and rack. Superimposed on the drift is a low-frequency oscillation, which correlates with the pinion revolutions. The predominant sources for these are concentric and axial run-out of the pinion as well as angular errors of the planetary gearbox. The third error component is a higher-frequency oscillation, that correlates with the tooth meshing frequency of the pinion. This is caused by deviations of the tooth flank geometry from the ideal involute, which are the result of manufacturing tolerances and also intentionally applied profile modifications [18], [19].

In contrast to the first two error components, the amplitude of these deviations decreases with increasing load torque. This change is due to the deformation of the tooth flanks, whose contact surface expands from a line in the no-load state to a lens-shaped area under load [5]. To improve the characteristics

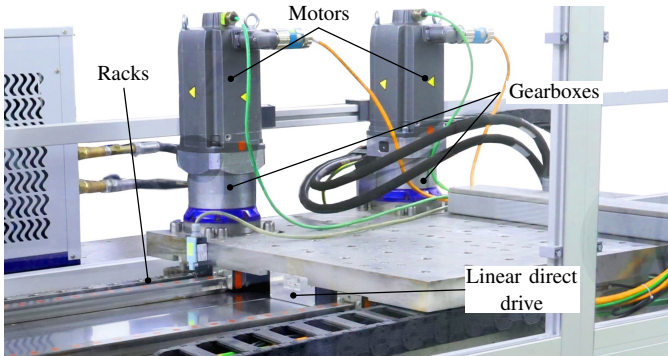


Fig. 2: Test bench with electrically preloaded RPDs, used for measurements.

under load, profile modifications can be applied in the design of the gearing, although this leads to higher TEs for lower loads [20]. In the following, the no-load error curves are referred to as geometric transmission errors TE_{geo} , while the superimposed deformations under load TE_{def} are referred to as quasi-static load-induced elastic errors [21].

While the general characteristics of the TE curves are comparable for both directions, the manufacturing tolerances in combination with a direction-dependent sign of the axial force on the pinion shaft, caused by the helical gearing [22], lead to direction-dependent alterations.

IV. IMPLICATIONS OF TRANSMISSION ERRORS FOR ELECTRICALLY PRELOADED SYSTEMS

As stated in Sec. I, the TEs have a negative effect on the path accuracy of the position-controlled system. This is investigated in the following subsection, after which the implications of the TEs for the preload control are examined.

A. Path accuracy

The TEs represent a deviation of the real system from the theoretical nominal transmission ratio i_n , which is applied to convert the translational output velocity of the position controller into the rotational target velocity of the motors, as illustrated in Fig. 1. Although the axis position is fed back by the linear measuring system, the deviations cannot be completely suppressed due to the limited bandwidth of the position control [1], [8]. For single drives, the theoretical relationship between TEs and the path error was derived in previous work [7]. However, for electrically preloaded systems, the path errors result from a superposition of the TEs of the two drivetrains. In the following, the theoretical relationship between the TEs and the path errors is derived for electrically preloaded systems and validated through measurements on the test bench.

As illustrated in Fig. 1, the mechanics of an electrically preloaded system can be sufficiently modeled as a three-mass oscillator [13], [14]. Under the common assumption of two

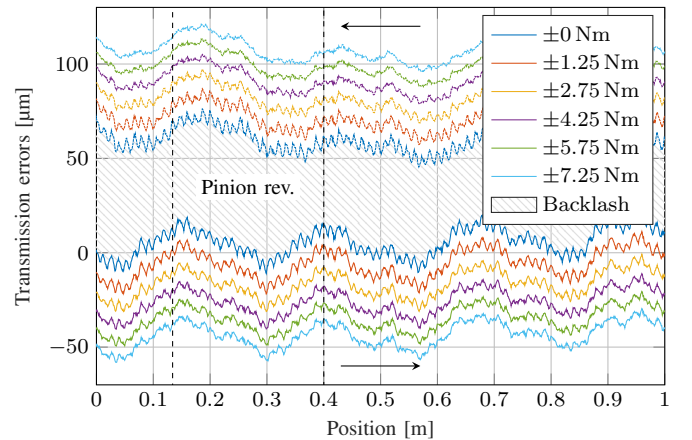


Fig. 3: TE curves of drive 2 for different preload torques.

identical drivetrains with identical stiffnesses, the table position is equal to the mean value of the translational positions of the two motors plus a superimposed deformation due to possible external forces. Since the TEs of the two drivetrains are added to the motor positions according to (1), they are superimposed to form a total error

$$TE_s = \frac{TE_1 + TE_2}{2}, \quad (2)$$

that likewise corresponds to their mean value in their effect on the overall system. Fig. 4 shows the individual and the superimposed TEs for a exemplary motion in positive direction with nominal preload torque.

Based on this, the relationship to the path errors of the system can be derived. The aim is to provide a practical approach for estimating the expected errors through reasonable simplifications without the need for extensive system identification as described in [12], [14]. Therefore, some simplifications are made. Since the bandwidth of the torque control is independent of the mechanical system and therefore considerably higher than that of the superimposed control loops, its transmission behavior is neglected [3]. Under the condition of a well-damped characteristic of the velocity control loops, it can also be assumed that the elasticities of the system are not excited [13]. As a result, with regard to the position control, the inertia of the drives and the table can be combined to form a total inertia J_s , which is accelerated by the drive torques. The preload control can be excluded from this analysis, as the velocity offset it provides for both drives is balanced out in total. As a consequence, given identical drivetrains, the two velocity control loops can be aggregated into a single one whose output torque is doubled. Fig. 5 shows the schematic of the system derived in this way. The TEs are modeled as an output disturbance between the motor and the table position according to (1) and (2).

Subsequently, based on the transfer function of the open position control loop

$$G_{pc,OL} = \frac{x_T}{x_d} \Big|_{TE_s=0} = \frac{2K_v K_p (T_n s + 1)}{s(J_s T_n s^2 + 2K_p T_n s + 2K_p)} \quad (3)$$

of this structure, the disturbance transfer function

$$G_{d,CL} = \frac{e_x}{TE_s} = -\frac{\frac{J_s T_n}{2K_p} s^3 + T_n s^2 + s}{\frac{J_s T_n}{2K_p} s^3 + T_n s^2 + (1 + K_v T_n) s + K_v} \quad (4)$$

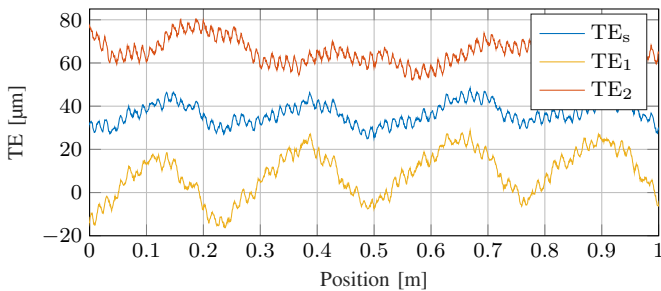


Fig. 4: Individual and superimposed TEs using (2).

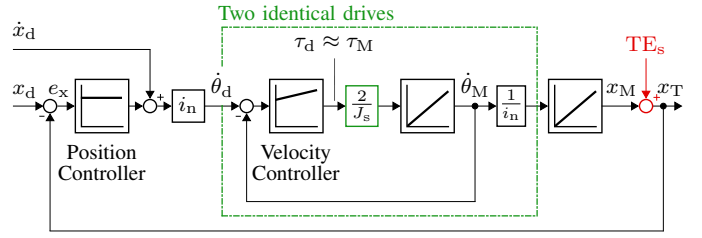


Fig. 5: Schematic of the simplified position controlled system with TEs as output disturbance.

from the total TE to the position error of the position control can be determined.

Analogous to the analysis for single drives in [7], this transfer function exhibits a high-pass characteristic, whose disturbance suppression is defined by the gain of the position controller K_v . Fig. 6 shows a comparison of the calculated and measured path errors for different velocities under nominal preload torque and no external load. A high level of concordance is evident, where primarily the high-frequency components of the TEs cannot be fully suppressed by the position control due to the limited bandwidth. Since the temporal frequency of the position-dependent TEs is increasing with the velocity, the path errors are also increasing.

B. Preload control

In addition to the impact on the path accuracy, the TEs also affect the precision of the preload control. Fig. 7 shows the control error of the torque balancing control e_{pl} for different velocities. For enhanced illustration, the torques were denoised with a moving average filter with a window length of 10 ms. Periodic deviations are evident, the amplitudes of which increase with the velocity.

The control error

$$e_{pl} \propto \frac{d}{dt} \frac{TE_1 - TE_2}{2} = \frac{d}{dt} TE_d \quad (5)$$

is proportional to the derivative of the averaged difference TE_d between the TEs of the two drivetrains, which is additionally depicted in Fig. 7 for a velocity of 100 mm/s. Again, this can be explained based on the modeling of the mechanics as

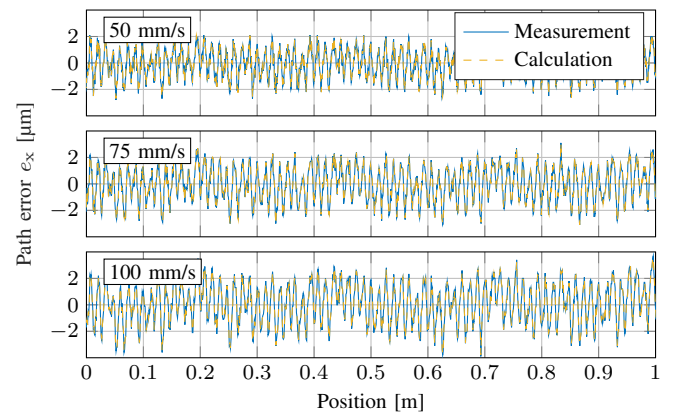


Fig. 6: Comparison of measured and calculated path errors using (4).

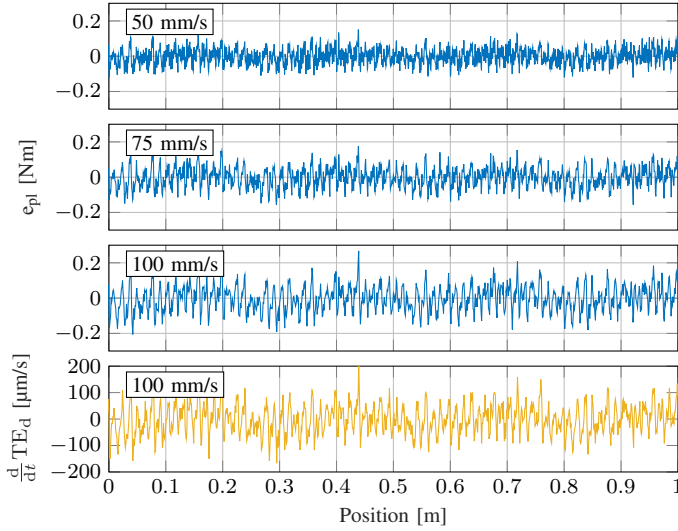


Fig. 7: Control error of the preload control for different velocities.

a three-mass oscillator. Under the assumption of a linearized stiffness around the operating point, the preload is proportional to the position difference between the two drives, which is influenced by the TEs. The derivation of the error difference results from the implementation of the torque balancing control at the velocity control level. The deduction of a complete transfer function analogous to (4) is omitted in this case, as this requires comprehensive modeling and, in particular, identification of the nonlinear stiffness and damping.

In summary, the investigations show that the TEs in the drivetrains of electrically preloaded systems have a negative impact on the path accuracy and the accuracy of the preload torque due to the limited bandwidths of the controllers. The path errors that occur can be estimated using (4) based on the mean value of the TEs of the two drivetrains. In contrast, the deviations of the preload torque correlate with the difference of the TEs.

V. TRANSMISSION ERROR COMPENSATION

To address this problem, this section presents a TE compensation for electrically preloaded systems. Therefore, an approach to model the errors is introduced first, before the implementation of an error compensation in the existing control structure is explained.

A. Error modeling

The approach utilized for error modeling corresponds to that for single drives developed in prior work [7], which is briefly explained in the following. For the electrically preloaded system, the TEs of both drives are modeled independently of each other. This allows for flexible adaptation to varying external loads and preload torques. The basis for the modeling are the measurements carried out in Sec. III to identify the load-dependent TEs through variation of the preload torque. Such a series of measurements can be carried out after assembly of the feed axis as part of the commissioning of the drive system.

The no-load geometric transmission errors TE_{geo} are stored in a lookup table. It was shown that an error compensation based purely on this data does not achieve a robust increase in accuracy and can even worsen it [9].

For this reason, the deformations under load TE_{def} , which represent the difference between the error curves under load and TE_{geo} , must also be taken into account. As mentioned above, these vary depending on manufacturing and assembly tolerances and are therefore highly individual and position-dependent. A two-stage machine learning approach is used to model these errors. The data pre-processing involves the interpolation of all measurement data with a step size of $50 \mu m$ and smoothing the curves through low-pass filters. To simplify the regression and improve performance, indirect training data is added in the form of the current gear meshing state M_j between rack and pinion, which is represented by a bump function for each tooth of the pinion j [9]. A 90/10 split between training and validation data is used. The developed two-stage modeling approach, whose structure is shown in Fig. 8, aims to address the bias-variance tradeoff and to optimize both generalization and accuracy [23]. First, periodic load-dependent deformations over a pinion revolution are modeled by a neural network (NN) that receives only the motor torque and the gear meshing state as input. This way, the NN models the general deformation characteristics of the gearing and thus enables generalization to unknown conditions. Since the axis position is not an input of the NN, localized effects are ignored. These nonperiodic residual errors are subsequently modeled by training a faster second model in the form of bagged regression trees (BRTs) [24], [25]. In this step, a highly accurate representation of the observed errors is prioritized over generalization capabilities. The outputs of both models are then summed to the predicted deformation $TE_{def,p}$ and, combined with TE_{geo} from the lookup table, result in the predicted total error TE_p .

In the previous work, it was demonstrated that this tailored approach for modeling the TEs can effectively mitigate the trade-off between accuracy and generalization [7]. Further advantages include the possibility of transfer learning of the NNs using pre-trained models of other drives or simulation data as well as countering wear-induced model degradation by updating the fast trainable BRTs [25], [26].

B. Control structure

To improve the path accuracy, the control structure introduced in Fig. 1 is extended by an error compensation using

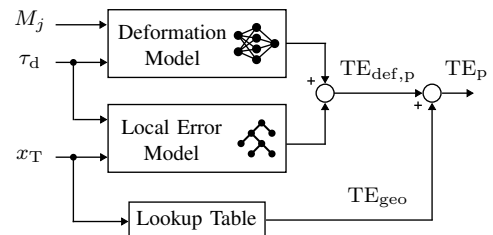


Fig. 8: Schematic architecture to model the state-dependent TEs.

the modeling described above. The implementation is based on the basic concept of previous work, which is extended to electrically preloaded systems following the findings in Sec. IV [7]. The underlying approach is to apply correction velocities for the position and preload control, which compensate for the deviations from the nominal transmission ratio caused by the TEs.

For this purpose, the TEs of both drives are calculated for the current position $x_{d,n}$ and the fed back motor torques $\tau_{d,1/2}$. Assuming a quasi-static load, the TEs are also predicted for the position $x_{d,n+1}$ in the next time step, which is obtained from the trajectory look ahead. The sample time $t_{s,c}$ of the error compensation is set to 5 ms. The superimposed error for both time steps is obtained using (2). The correction velocity for the position control loop

$$v_{c,pc} = \frac{TE_{s,n+1} - TE_{s,n}}{t_{s,c}} \quad (6)$$

equals the difference between these two errors divided by the sampling time. This velocity is subtracted from the target velocity output by the position controller, as illustrated in Fig. 9.

As derived in Sec. IV through (5), in contrast to the path errors, the control errors of the preload torque result from the difference between the TEs of both drives. Consequently, the angular correction velocity for the preload control loop

$$\dot{\theta}_{c,pl} = i_n \frac{TE_{d,n+1} - TE_{d,n}}{t_{s,c}} \quad (7)$$

is calculated as the change of TE_d between the current and the next time step. As the preload control operates at the velocity control level, the correction velocity is multiplied by the nominal transmission ratio. The resulting velocity is subtracted from the output of the preload control, as illustrated in the schematic.

The compensation operates through a feedforward path that continuously calculates the correction velocities based on the planned trajectory. While the target motor torques of both drives are fed back, they only serve as state variables for the error models. This way, their impact on the correction velocity is strictly constrained and there is no direct feed-through

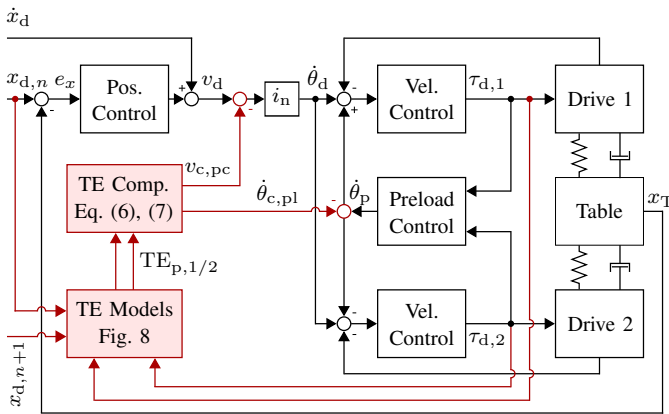


Fig. 9: Control scheme with TE compensation.

to the compensation output. Consequently, the parameters and stability characteristics of the position control loop are retained.

VI. EXPERIMENTAL VALIDATION

To validate the effectiveness of the error compensation, the control structure depicted in Fig. 9 is implemented on the test system introduced in Sec. III and the improvement in path accuracy achieved is evaluated and quantified. To examine different operating conditions, various velocities and external loads are considered. The external loads are applied by the redundant linear direct drive (LDD) installed on the test bench, the preload torque is set to the nominal value during all tests and is not varied. Only the results for the positive direction of travel are presented in the following, the negative direction is comparable.

Fig. 10 shows an exemplary comparison of the path and preload errors with and without TE compensation for a motion with a velocity of 100 mm/s and an external load of 2500 N applied by the LDD. A significant improvement of the cumulative absolute path error by 59.7 % as well as a reduction of the cumulative absolute error of the preload torque by 46.1 % are observed.

Fig. 11 illustrates the improvement of the mean absolute path error for the entire examined operating range. For this purpose, the travel range was cycled repeatedly at a constant velocity, with the external load being increased by 500 N between each cycle. Each cycle was carried out once without and once with active TE compensation. The percentage improvement in the mean absolute path error achieved by the compensation was calculated in sections of 100 mm and plotted as a surface over load and position. The entire measurement procedure was carried out for three different velocities. Overall, a robust performance of the compensation is evident. The improvement decreases for higher loads and lower velocities. However, this is not attributable to the effectiveness of the compensation, but rather to the lower amplitude of high-frequency TEs for higher loads and the lower path errors for lower velocities in general, as discussed in Sec. III and IV. On

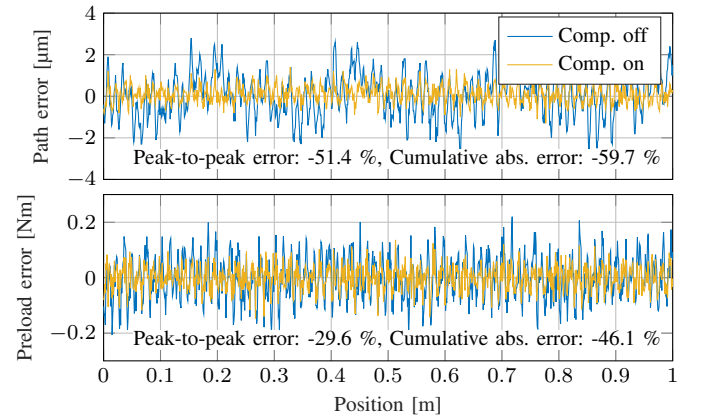


Fig. 10: Comparison of the path and preload error with and without compensation for a velocity of 100 mm/s and 2500 N external load.

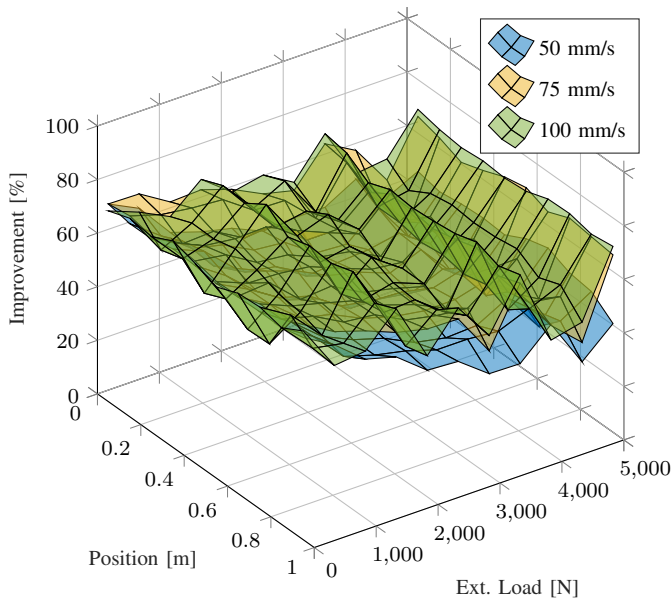


Fig. 11: Improvement of path accuracy by the compensation for varying external loads and velocities.

average, for a velocity of 50 mm/s an improvement of 49.9 %, for 75 mm/s of 61.8 % and for 100 mm/s of 61.6 % can be achieved.

VII. CONCLUSION

This paper presents a novel approach for the compensation of transmission errors in feed axes with electrically preloaded rack-and-pinion drives. Tolerances during production and assembly as well as the elastic deformation of the gearing lead to deviations between the rotation of the motor and the translational motion of the axis. Due to the limited bandwidth of the position control, these deviations cannot be entirely suppressed despite the use of linear measuring systems. Based on the system structure, the theoretical relationship between the transmission errors of the two drivetrains of electrically preloaded systems and the resulting path errors is derived and experimentally validated. Furthermore, the negative impact of the transmission errors on the control performance of the preload control is derived and investigated.

Based on the knowledge gained, an error compensation is developed and integrated into the existing drive control. For this purpose, a machine learning approach to error modeling for individual drives from prior work is utilized and adapted for electrically preloaded systems. The error models are trained by varying the preload torque during commissioning of the system. The models are then applied to predict the state-dependent transmission errors during operation and compensate for them with feedforward signals for the position and preload control.

The effectiveness of the error compensation is validated through experimental investigations on a test bench. Thereby the robustness of the developed approach is demonstrated for different velocities and external loads. Across the examined

operating range, an average reduction of the path errors of 57.8 % can be observed.

Future work includes the development of a performance monitoring of the error models for automatic adaptation to wear and the evaluation of the approach in machining processes.

REFERENCES

- [1] Y. Altintas, A. Verl, C. Brecher, L. Uriarte, and G. Pritschow, "Machine tool feed drives," *CIRP Annals*, vol. 60, no. 2, pp. 779–796, 2011.
- [2] L. Uriarte *et al.*, "Machine tools for large parts," *CIRP Annals*, vol. 62, no. 2, pp. 731–750, 2013.
- [3] C. Brecher and M. Weck, *Machine Tools Production Systems 3*. Wiesbaden: Springer, 2022.
- [4] L. Steinle, A. Lechler, M. Neubauer, and A. Verl, "Experimental investigation into the implications of transmission errors for rack-and-pinion drives," *Production Engineering*, vol. 16, no. 4, pp. 453–463, 2022.
- [5] V. Vullo, *Gears: Volume 1: Geometric and Kinematic Design*, 1st ed. Cham: Springer, 2020, vol. 10.
- [6] T. Engelberth, S. Apprich, J. Friedrich, D. Coupek, and A. Lechler, "Properties of electrically preloaded rack-and-pinion drives," *Production Engineering*, vol. 9, no. 2, pp. 269–276, 2015.
- [7] L. Steinle, V. Leipe, A. Lechler, and A. Verl, "Learning Compensation of the State-Dependent Transmission Errors in Rack-and-Pinion Drives," in *2024 European Control Conference (ECC)*, 2024, pp. 2441–2447.
- [8] D. Lyu, Q. Liu, H. Liu, and W. Zhao, "Dynamic error of CNC machine tools: a state-of-the-art review," *The International Journal of Advanced Manufacturing Technology*, vol. 106, no. 5-6, pp. 1869–1891, 2020.
- [9] A. Verl and L. Steinle, "Adaptive compensation of the transmission errors in rack-and-pinion drives," *CIRP Annals*, vol. 71, no. 1, pp. 345–348, 2022.
- [10] T. Engelberth, "Adaptive Verspannung von Zahnstange-Ritzel-Antrieben [Adaptive Preloading of Rack-and-Pinion Drives]," Ph. D. thesis, Universität Stuttgart, 2019.
- [11] L. Steinle, "Replication Data for: Compensation of the Transmission Errors in Electrically Preloaded Rack-and-Pinion Drives," <https://doi.org/10.18419/DARUS-5160>, DaRUS, 2025.
- [12] A. Verl and T. Engelberth, "Adaptive preloading for rack-and-pinion drive systems," *CIRP Annals*, vol. 67, no. 1, pp. 369–372, 2018.
- [13] O. Zirn, "Machine tool analysis: modelling, simulation and control of machine tool manipulators," Habilitation thesis, ETH Zurich, 2008.
- [14] V. Leipe, L. Steinle, X. A. Bikuña, A. Lechler, and A. Verl, "Modeling and Identification of Load-Dependent Properties for Electrically Preloaded Rack-and-Pinion Drives," in *2024 European Control Conference (ECC)*, 2024, pp. 2460–2466.
- [15] Dr. Johannes Heidenhain GmbH, "Dynamic Precision – Machining Dynamically and with High Accuracy," 2013, Accessed: Apr. 23, 2025. [Online]. Available: <https://www.heidenhain.us/wp-content/uploads/2022/08/DynamicPrecision.pdf>
- [16] L. Steinle, V. Leipe, A. Lechler, and A. Verl, "Automatic preload adaptation for rack-and-pinion drives to maximize performance and energy efficiency," *Production Engineering*, vol. 19, no. 3-4, pp. 419–427, 2025.
- [17] Siemens AG, "Drive optimization guide: V2.1," 2019, Accessed: Apr. 23, 2025. [Online]. Available: <https://publikacje.siemens-info.com/pdf/677/>
- [18] I. Ulacia, M. B. Sánchez, A. Iñurritegui, A. Arana, J. Larrañaga, and J. I. Pedrero, "Theoretical analysis of transmission error in rack and pinion systems," *MATEC Web of Conferences*, vol. 387, p. 01001, 2023.
- [19] M. Benatar, M. Handschuh, A. Kahraman, and D. Talbot, "Static and Dynamic Transmission Error Measurements of Helical Gear Pairs With Various Tooth Modifications," *Journal of Mechanical Design*, vol. 141, no. 10, p. 103301, 2019.
- [20] U. Kissling, "Effects of profile corrections on peak-to-peak transmission error," *Gear technology*, vol. 27, no. 5, pp. 52–61, 2010.
- [21] W. Gao *et al.*, "Machine tool calibration: Measurement, modeling, and compensation of machine tool errors," *International Journal of Machine Tools and Manufacture*, vol. 187, p. 104017, 2023.

- [22] V. Roda-Casanova and F. Sanchez-Marin, "Contribution of the deflection of tapered roller bearings to the misalignment of the pinion in a pinion-rack transmission," *Mechanism and Machine Theory*, vol. 109, pp. 78–94, 2017.
- [23] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. New York, NY: Springer, 2009.
- [24] L. Breiman, "Bagging predictors," *Machine Learning*, vol. 24, no. 2, pp. 123–140, 1996.
- [25] W.-Y. Loh, "Classification and regression trees," *WIREs Data Mining and Knowledge Discovery*, vol. 1, no. 1, pp. 14–23, 2011.
- [26] C. C. Aggarwal, *Neural Networks and Deep Learning*. Cham: Springer, 2018.